

Podman: A Powerful Alternative to Docker

With its Docker-compatible CLI and daemon-less architecture, Podman is an excellent choice for those looking to adopt a secure, scalable, and flexible approach to container management.



In modern software development, containerisation has become essential for building, testing, and deploying applications. Containers allow applications to run consistently across various computing environments, providing isolation, portability, and flexibility. This is achieved by packaging the code, runtime, system libraries, and dependencies into a container unit.

While Docker has long been the dominant containerisation tool, a powerful alternative has emerged: Podman. Short for Pod Manager, this tool is designed for managing OCI (Open Container Initiative)-

compliant containers and images. As a Docker alternative, Podman offers more security, flexibility, and the ability to run containers without requiring a central daemon. This daemon-less architecture is one of its key differentiators, making Podman particularly appealing for users focused on security and user control.

Also, Podman maintains compatibility with Docker images and commands. The Podman CLI mimics the Docker CLI, meaning you can replace *docker* with *podman* in many commands. This makes transitioning from Docker to Podman seamless for developers.

Core features of Podman

Podman offers a range of features that provide container management flexibility, security, and scalability.

Daemon-less operation: Podman does not rely on a running daemon to manage containers. This improves system security since there's no long-running root-privileged service required. Containers are started and managed directly by the user.

Rootless containers: A standout feature, Podman allows users to run containers without root access. By avoiding root permissions, security risks are minimised, making Podman ideal for multi-user environments or